

Assignment 12.

1. Set up a new Firebase project in the Firebase console, enable Firestore, and connect it to your React app using the firebase npm package.

2. Create a new Firestore collection called 'restaurants' and build a simple React form that allows users to add a restaurant with fields: name and cuisine. On form submit, add the data as a new document in the 'restaurants' collection.

3. Fetch all documents from the 'restaurants' collection and display each restaurant's name and cuisine in a list on your React page.

***Hint:** Use Firestore's getDocs() method to retrieve the data.*

4. Build a filter feature for your restaurant list: add a text input that lets users type a cuisine (like 'Italian' or 'Chinese'), and only show restaurants that match the entered cuisine in real-time.

Answer:- 1.

Firebase.jsx

```
import { initializeApp } from "firebase/app";
import { getAuth } from "firebase/auth";

const firebaseConfig = {
  apiKey: "AIzaSyDuzrblSgnCQ_n0HJUKcYtPNz-1k5rsobo",
  authDomain: "react-firebase-3a941.firebaseio.com",
  projectId: "react-firebase-3a941",
  storageBucket: "react-firebase-3a941.firebaseio.com",
  messagingSenderId: "744439850234",
  appId: "1:744439850234:web:8a54a8b2c01532b111d543",
  measurementId: "G-QYESDFS1GQ"
};

const app = initializeApp(firebaseConfig);

export const auth = getAuth(app);

export default app;
```

Step 3: Enable Firestore

1. Open Firebase Console.
2. Create a new project.
3. Click **Firestore Database**.
4. Click **Create Database**.

5. Start in **Test Mode**.
6. Create the database.

App.jsx

```
import { useEffect, useState } from "react";
import { db } from "../firebase";

import { collection, addDoc, getDocs } from "firebase/firestore";

function App() {
  const [name, setName] = useState("");
  const [cuisine, setCuisine] = useState("");

  const [restaurants, setRestaurants] = useState([]);
  const [filterCuisine, setFilterCuisine] = useState("");

  // Add Restaurant
  const handleSubmit = async (e) => {
    e.preventDefault();

    if (!name || !cuisine) {
      alert("Please fill all fields");
      return;
    }

    try {
      await addDoc(collection(db, "restaurants"), {
        name,
        cuisine,
      });

      setName("");
      setCuisine("");

      fetchRestaurants();
    } catch (error) {
      console.log(error);
    }
  };

  // Fetch Restaurants
  const fetchRestaurants = async () => {
    try {
      const querySnapshot = await getDocs(collection(db, "restaurants"));

      const data = querySnapshot.docs.map((doc) => ({
```

```

        id: doc.id,
        ...doc.data(),
      }));

      setRestaurants(data);
    } catch (error) {
      console.log(error);
    }
  };

  useEffect(() => {
    fetchRestaurants();
  }, []);

  // Filter Logic
  const filteredRestaurants = restaurants.filter((restaurant) =>
    restaurant.cuisine.toLowerCase().includes(filterCuisine.toLowerCase()),
  );

  return (
    <div
      style={{
        maxWidth: "600px",
        margin: "20px auto",
        fontFamily: "Arial",
      }}
    >
      <h1>Restaurant Manager</h1>

      { /* Add Restaurant Form */ }
      <form onSubmit={handleSubmit}>
        <input
          type="text"
          placeholder="Restaurant Name"
          value={name}
          onChange={(e) => setName(e.target.value)}
        />

        <br />
        <br />

        <input
          type="text"
          placeholder="Cuisine"
          value={cuisine}
          onChange={(e) => setCuisine(e.target.value)}
        />

```

```

    <br />
    <br />

    <button type="submit">Add Restaurant</button>
  </form>

  <hr />

  { /* Filter Input */}
  <input
    type="text"
    placeholder="Filter by Cuisine"
    value={filterCuisine}
    onChange={(e) => setFilterCuisine(e.target.value)}
  />

  <hr />

  { /* Restaurant List */}
  <h2>Restaurants</h2>

  {filteredRestaurants.length === 0 ? (
    <p>No Restaurants Found</p>
  ) : (
    <ul>
      {filteredRestaurants.map((restaurant) => (
        <li key={restaurant.id}>
          <strong>{restaurant.name}</strong>
          { " - " }
          {restaurant.cuisine}
        </li>
      ))}
    </ul>
  )}
</div>
);
}

export default App;

```